

Differentiation

Week 3C

Overview

Techniques of Derivatives

Key Ideas - Rule 4: Quotient Rule

- Rule 5: Chain Rule

- Derivative of Powers of x Combining Rule 1-5

Key Idea – Rule 4: Quotient Rule

If $u = f(x)$ and $v = g(x)$ are differentiable functions, then $\frac{u(x)}{v(x)}$ can be differentiated using the quotient rule:

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{\frac{du}{dx}v + u\frac{dv}{dx}}{v^2} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

In simplified form we could write

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$$

Suggestion: Learn the quotient rule independently of the variable:

“Derivative of the top times the bottom minus the top times the derivative of the bottom all over the bottom squared.”

Example:

Differentiate $f(x) = \frac{4x^3}{x^2 + 1}$ using the quotient rule.

$$u = 4x^3$$

$$v = x^2 + 1$$

$$u' = 12x^2$$

$$v' = 2x$$

$$y' = \frac{u'v - uv'}{v^2}$$

$$= \frac{12x^2 \times (x^2 + 1) - 4x^3 \times 2x}{(x^2 + 1)^2}$$

$$= \frac{12x^4 + 12x^2 - 8x^4}{(x^2 + 1)^2}$$

$$= \frac{4x^4 + 12x^2}{(x^2 + 1)^2}$$

Note: Leave the denominator in factorised form but expand and simplify the numerator.

Exercise:

Use the quotient rule to differentiate the following function:

$$y = \frac{2x+1}{x^3-x}$$

$$y = \frac{u}{v}$$

$$u = 2x+1 \quad v = x^3-x$$

$$u' = 2$$

$$v' = 3x^2-1$$

$$\frac{dy}{dx} = \frac{u'v - uv'}{v^2}$$

$$\frac{dy}{dx} = \frac{2 \times (x^3-x) - (2x+1) \times (3x^2-1)}{(x^3-x)^2}$$

$$= \frac{2x^3 - \cancel{2x} - 6x^3 + \cancel{2x} - 3x^2 + 1}{(x^3-x)^2}$$

$$= \frac{-4x^3 - 3x^2 + 1}{(x^3-x)^2}$$

$$\frac{-4x^3 - 3x^2 + 1}{(x^3-x)^2}$$

Exercise:

Given the function $y = \frac{2}{2x+1}$ evaluate $y'(1)$.

$$y = \frac{u}{v} \quad y' = \frac{u'v - uv'}{v^2}$$

$$u = 2 \quad v = 2x + 1$$

$$u' = 0 \quad v' = 2$$

$$y' = \frac{0 \times (2x+1) - 2 \times 2}{(2x+1)^2}$$

$$y' = \frac{0 - 4}{(2x+1)^2} = \frac{-4}{(2x+1)^2}$$

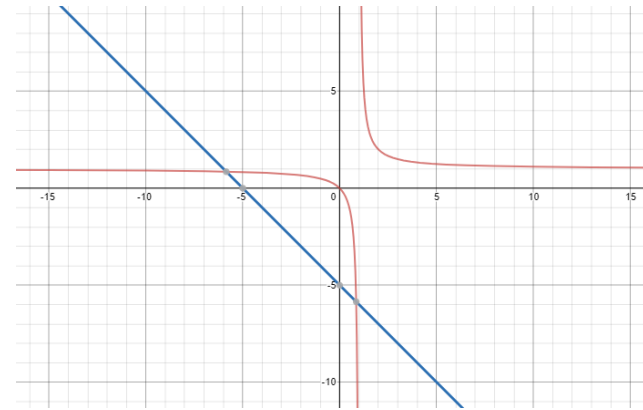
$$\begin{aligned} y'(1) &= \frac{-4}{(2 \times 1 + 1)^2} \\ &= \frac{-4}{3^2} \\ &= \frac{-4}{9} \end{aligned}$$

Note: The derivative for this example could have been found using the chain rule (shown next) instead of the quotient rule.

$$u = x \quad u' = 1 \quad v = x - 1 \quad v' = 1$$

Exercise:

At what points on the curve $y = \frac{x}{x-1}$ is the tangent parallel to the line $x + y + 5 = 0$?



$$1. \text{ gradient of } x + y + 5 = 0 \\ y = -x - 5 \therefore m = -1$$

$$2. \text{ find } y' = \frac{1 \times (x-1) - x \times 1}{(x-1)^2}$$

5. points are

$$(0, 0), (2, 2)$$

$$= \frac{x-1-x}{(x-1)^2} \\ = \frac{-1}{(x-1)^2}$$

$$3. \text{ now let } y' = m$$

$$\frac{-1}{(x-1)^2} = -1 \\ -1 = -1(x-1)^2 \\ 1 = (x-1)^2 \\ \pm 1 = x-1$$

$$x = \pm 1 + 1 = 0, 2$$

Note: For the above exercise we showed the graph so that we can visualise the question. You will not be given the graph in an exam.

$$\bullet y\text{-coord: } x = 0 \Rightarrow y = 0$$

$$x = 2 \Rightarrow y = 2$$

Key Idea – Rule 5: Chain Rule

Another method that can be used to find the derivative of a composite function is often referred to as a “function of a function”. We call this method the **chain rule**.

Review of a Composite Function

We can find the **composition** of two functions by placing one inside the other. Composite functions are sometimes called “function of functions”. For two functions, $f(x)$ and $g(x)$, this is denoted by $(f \circ g)(x) = f(g(x))$.

Examples:

If $f(x) = x^5$ and $g(x) = 2x + 3$

then $f(g(x)) = (2x + 3)^5$ ie $(2x + 3)^5$ is a composite function.

If $f(x) = x^8$ and $g(x) = x^2 - 1$

then $f(g(x)) = (x^2 - 1)^8$ ie $(x^2 - 1)^8$ is a composite function.

Note: The following differentiation rule is the most important general rule for both differentiation (and later for anti-differentiation).

If $y = f(x)$ and $u = g(x)$ are two differentiable functions,

then $y = f(g(x)) = f(u)$ is a differentiable function and we can use the chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

In simplified form we could write

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \times g'(x)$$

Suggestion: Learn the chain rule independently of the variable:

“Differentiate the outside function first leaving the inside the same, then multiply by the derivative of the inside function.”

When using the chain rule follow these steps below:

1. Identify $u = g(x)$ and find $\frac{du}{dx}$
2. Identify $y = f(u)$ and find $\frac{dy}{du}$
3. $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

Example: Differentiate $y = f(x) = (x^2 + 1)^{10}$.

1. Let $u = g(x) = x^2 + 1$ and $\frac{du}{dx} = 2x$.
2. $y = f(u) = u^{10}$ and $\frac{dy}{du} = 10u^9$.
3.
$$\begin{aligned}\frac{d(y)}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= (10u^9)(2x) \\ &= 20xu^9 \\ &= 20x(x^2 + 1)^9\end{aligned}$$

Note: In this subject we will only use the simplified form which is consistent with DPEN141.

Example:

Differentiate $y = f(x) = (x^2 + 1)^{10}$.

We will now show you how to use chain rule independently of the variable by using

“Differentiate the outside function first leaving the inside the same, then multiply by the derivative of inside function.”

Derivative of outside function *Derivative of inside function*

$$y' = 10(x^2 + 1)^9 \times (2x)$$

Leave the inside the same

$$= 20x(x^2 + 1)^9$$

Example: If $y = \sqrt{5 - 6x}$, find $\frac{dy}{dx}$.

$$y = \sqrt{5 - 6x}$$

$$= (5 - 6x)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(5 - 6x)^{\frac{1}{2}-1}(-6)$$

$$= \frac{1}{2}(5 - 6x)^{-\frac{1}{2}}(-6)$$

$$= -3(5 - 6x)^{-\frac{1}{2}}$$

$$= \frac{-3}{\sqrt{5 - 6x}}$$

Exercise:

- (a) Find $\frac{dy}{dx}$ for $y = 3(9 - 2x)^{11}$.

$$\begin{aligned}\frac{dy}{dx} &= 3 \times 11 (9 - 2x)^{11-1} \times (-2) \\ &= 33 (9 - 2x)^{10} \times (-2) \\ &= -66 (9 - 2x)^{10}\end{aligned}$$

- (b) Evaluate the derivative of $y = (2x^3 + 3x - 1)^4$ at $x = 1$.

$$\begin{aligned}\frac{dy}{dx} &= 4 (2x^3 + 3x - 1)^3 \times (6x^2 + 3) \\ &= 12 (2x^2 + 1) (2x^3 + 3x - 1)^3\end{aligned}$$

$$\begin{aligned}\text{at } x = 1: \\ &12 (2 \times (-1)^2 + 1) (2 \times (-1)^3 + 3 \times (-1) - 1)^3 \\ &= 12 (3) (-2 - 3 - 1) \\ &= -216\end{aligned}$$

(c) Find the gradient function of $y = (x^3 + 3)^{\frac{1}{3}}$.

$$y' = \frac{1}{3} (x^3 + 3)^{\frac{1}{3} - 1} \times (3x^2)$$

$$y' = x^2 (x^3 + 3)^{-2/3}$$

$$= \frac{x^2}{\sqrt[3]{(x^3 + 3)^2}}$$

Exercise:

- (a) Find $f'(x)$ given $f(x) = \frac{2}{2x+1}$ (same example as shown previously in the quotient rule section)

rewrite with negative indices

$$f(x) = \frac{2}{2x+1} = 2(2x+1)^{-1}$$

$$f'(x) = -2(2x+1)^{-2} \times 2 = -4(2x+1)^{-2} = \frac{-4}{(2x+1)^2}$$

- (b) Evaluate $f'(1)$ if $f(x) = \frac{1}{\sqrt{2x-1}}$.

$$f(x) = \frac{1}{\sqrt{2x-1}} = (2x-1)^{-1/2}$$

$$f'(x) = -\frac{1}{2}(2x-1)^{-3/2} \times (2)$$

$$= -(2x-1)^{-3/2}$$

$$= \frac{-1}{\sqrt{(2x-1)^3}}$$

Key Idea – Derivative of Powers of x Combining Rule 1-5

We now combine the power rule with the product rule, quotient rule and chain rule in order to differentiate more complex expressions built from powers of x .

Exercise: [Chain Rule + Quotient Rule]

Differentiate $y = \frac{2x}{(1+3x^3)^2}$.

$$y = \frac{u}{v}$$

$$y' = \frac{u'v - uv'}{v^2}$$

$$u = 2x$$

$$u' = 2$$

$$v = (1+3x^3)^2$$

$$v' = 2(1+3x^3)' \times 3x^2$$
$$= 18x^2(1+3x^3)$$

$$\frac{dy}{dx} = \frac{2 \times (1+3x^3)^2 - 2x \times (18x^2(1+3x^3))}{(1+3x^3)^2}$$

$$= \frac{2(1+3x^3) [(1+3x^3) - 18x^3]}{(1+3x^3)^2}$$

$$= \frac{2[1-15x^3]}{(1+3x^3)}$$

Exercise: [Chain Rule + Product Rule]

Find the gradient of the tangent to the curve $y = (x+1)(1-x^2)^3$ at the point $(1, 0)$.

$$y = u \times v$$

$$y' = u'v + uv'$$

$$u = (x+1)$$

$$u' = 1$$

$$v = (1-x^2)^3$$

$$v' = 3(1-x^2)^2 \times (-2x)$$

$$= -6x(1-x^2)^2$$

$$y' = 1 \times (1-x^2)^3 + (x+1) \times (-6x(1-x^2)^2)$$

gather like terms

$$y' = (1-x^2)^2 [(1-x^2) - 6x(x+1)]$$

$$y' = (1-x^2)^2 [1-x^2-6x^2-6x]$$

$$y' = (1-x^2)^2 [1-7x^2-6x]$$

gradient at $(1, 0)$

gradient at $(1, 0)$

sub $x=1$ into

$$y' = (1-x^2)^2 [1-7x^2-6x]$$

$$y'(1) = (1-1^2)^2 [1-7 \times 1^2 - 6 \times 1]$$

$$= 0 [-4]$$

$$= 0$$

$$\therefore m = 0$$